Dynamic Entity Field Integration (DEFI): A Coupled Lagrangian Construction Unifying SEFI Identity and GWFM Geometry

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Dynamic Entity Field Integration (DEFI) has appeared throughout the SEFI/GWFM literature as the unifying field law binding the Single Entity Field Interpretation (SEFI) identity layer to the Geometric Worldline Field Model (GWFM) geometry layer, expressed through the invariance condition $\frac{D}{d\tau}(\kappa, \tau_g, \dot{I}^a, \ddot{I}^a) = 0$ on a stability manifold Σ. This condition has to date been stipulated rather than derived, a gap noted explicitly in the Limitations of the unified geometric field theory. This paper supplies the missing construction. DEFI is defined as a joint Lagrangian dynamical system on $M \times \mathcal{I}$, with a fully symmetric coupling potential linking GWFM curvature/torsion invariants to SEFI's Sovereignty and Warp identity coordinates, weighted by the Authorship coordinate. The stipulated invariance condition is recovered as the equilibrium condition of this coupled system rather than assumed. A conserved Origin charge is derived via Noether's theorem for the decoupled Origin sector. This construction requires an explicit extension of SEFI, promoting its originally scalar identity functional $F = O + A + S + W$ to a four-vector $I^a = (I_O, I_A, I_S, I_W)$ on an identity-space manifold $\mathcal{I} = \mathbb{R}^4$, which is stated here as a deliberate axiomatic extension rather than a silent reinterpretation.

I. INTRODUCTION

Across the SEFI/GWFM literature, Dynamic Entity Field Integration (DEFI) is invoked as "the unifying field law that binds SEFI identity and GWFM geometry into a single stability surface"[1,2], and its only concrete mathematical content is the stipulated dynamical invariance condition

$$\frac{D}{d\tau}\left(\kappa,\ \tau_g,\ \dot{I}^a,\ \ddot{I}^a\right) = 0 \qquad \text{on } \Sigma, \tag{1}$$

appearing in the unified geometric field theory[3]. No prior document derives Eq. (1) from an action principle; it is asserted as a definition of ideal evolution. The unified paper's own Limitations section acknowledges this directly: "the operator T is phenomenological and not derived from a variational principle." The present paper closes that gap by constructing DEFI as a genuine coupled dynamical system whose equations of motion reproduce Eq. (1) as an equilibrium condition, rather than as an axiom.

A. A Required Extension of SEFI

The original SEFI formal construction[1] defines field identity as a *scalar* functional,

$$F(t) = O(r(t)) + A(r(t), \kappa(t)) + S(r(t)) + W(r(t)), \tag{2}$$

summing the Origin, Authorship, Sovereignty, and Warp layers into a single number evaluated along the worldline. The later unified paper[3] instead treats $I^a = (I_O, I_A, I_S, I_W)$ as coordinates on an identity-space manifold $\mathcal{I} = \mathbb{R}^4$ with metric $g_I = \mathrm{diag}(\alpha, \beta, \gamma, \delta)$, differentiable in τ as a four-vector. These are not the same mathematical object: a scalar sum cannot be covariantly differentiated in four independent directions, so Eq. (1) is only well-defined under the vector reading.

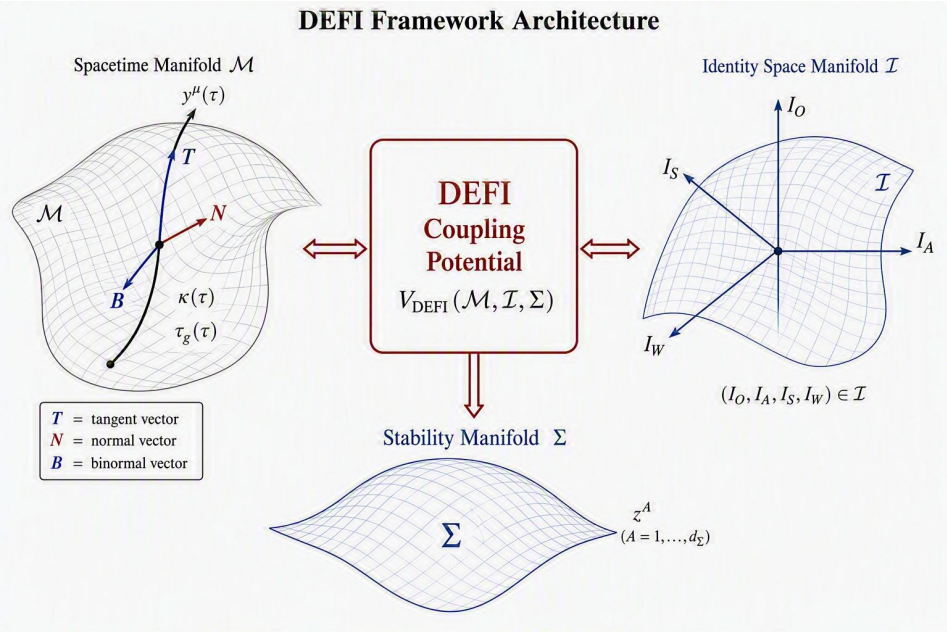


FIG. 1. Conceptual structure of DEFI as the variational bridge between SEFI identity-space dynamics and GWFM worldline geometry. The coupled DEFI framework generates the integrated consistency conditions that appear in the unified geometric field theory.

This paper adopts the vector reading explicitly, as a stated extension of SEFI rather than a silent correction of the earlier scalar definition. We record this as:

> **DEFI Axiom 0 (Vectorization of Identity).** SEFI's scalar identity functional $F = O + A + S + W$ is extended to a four-vector $I^a = (I_O, I_A, I_S, I_W) \in \mathcal{I}$ carrying an independent metric $g_I = \text{diag}(\alpha, \beta, \gamma, \delta)$. The original scalar F is recovered as $F = \delta_{ab}\, \mathbf{1}^a I^b$, the coordinate sum of the vector, so that Eq. (2) is a special contraction of the DEFI-extended identity vector rather than a competing definition.

II. CONFIGURATION SPACE

DEFI is defined on the product space $M \times \mathcal{I}$, where M is the spacetime manifold carrying the GWFM worldline and $\mathcal{I} = \mathbb{R}^4$ is the SEFI identity-space manifold of Axiom 0. A single trajectory

$$z(\tau) = (y^\mu(\tau),\ I^a(\tau)) \in M \times \mathcal{I} \tag{3}$$

carries both sectors simultaneously in one proper-time parameter τ. Geometric invariants $\kappa(\tau), \tau_g(\tau)$ are defined through the Frenet–Serret frame (T, N, B) of $y(\tau)$[7], exactly as in GWFM[2].

III. SECTOR LAGRANGIANS

A. Geometry Sector (GWFM, unchanged)

$$L_{\text{geom}} = -m\sqrt{-g_{\mu\nu}(y)\, \dot{y}^\mu \dot{y}^\nu}. \tag{4}$$

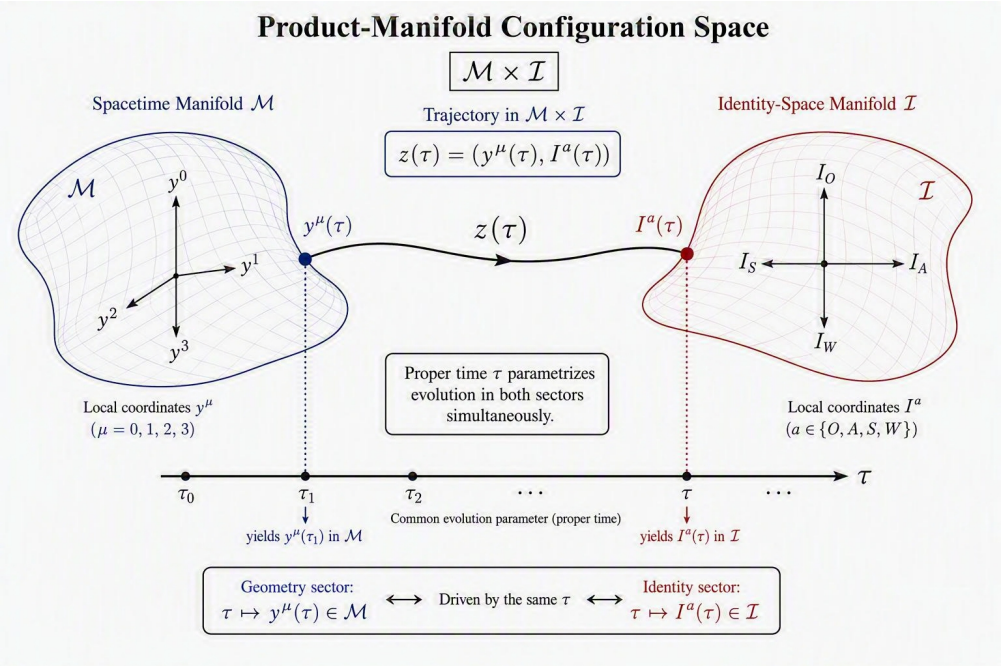


FIG. 2. Product-manifold configuration space $M \times \mathcal{I}$. The trajectory $z(\tau) = (y^\mu(\tau), I^a(\tau))$ evolves simultaneously through the spacetime manifold M and the identity-space manifold $\mathcal{I}$ under a common proper-time parameter τ.

B. Identity Sector

$$L_{\text{ident}} = \frac{1}{2} g_I^{ab} \dot{I}_a \dot{I}_b - \frac{1}{2} \left(\alpha I_O^2 + \beta I_A^2 + \gamma I_S^2 + \delta I_W^2 \right). \tag{5}$$

Each identity coordinate is, in isolation, a harmonic oscillator in τ, and the four coordinates together form a system of coupled oscillators once U_{DEFI} is introduced below[9]. This gives SEFI's Origin-layer stability bound $\|\ddot{r}\| \leq S_0$[1] a dynamical origin: bounded oscillatory motion is now a derived consequence of L_{ident} rather than an externally imposed inequality.

IV. THE DEFI COUPLING POTENTIAL

The coupling potential is required to be fully symmetric: geometry acts on identity and identity acts on geometry within a single stationary action, with neither sector prior to the other. We propose

$$U_{\text{DEFI}}(\kappa, \tau_g, I^a) = \mu_1(I_A)\, \kappa^2 I_S \; + \; \mu_2(I_A)\, \tau_g^2 I_W. \tag{6}$$

This pairing is not arbitrary; each term is fixed by the canonical SEFI layer definitions[1]:

- **Sovereignty ↔ curvature.** SEFI defines Sovereignty through the stability/acceleration bound of Eq. (2)'s origin condition; curvature κ is GWFM's geometric measure of how sharply a worldline accelerates. Pairing I_S with κ^2 makes this correspondence dynamical.
- **Warp ↔ torsion.** The SEFI paper states explicitly that warp operators perform "curvature modulation, torsion modulation"[1]; torsion τ_g is GWFM's natural analogue of twisting transformation. Pairing I_W with τ_g^2 follows directly.

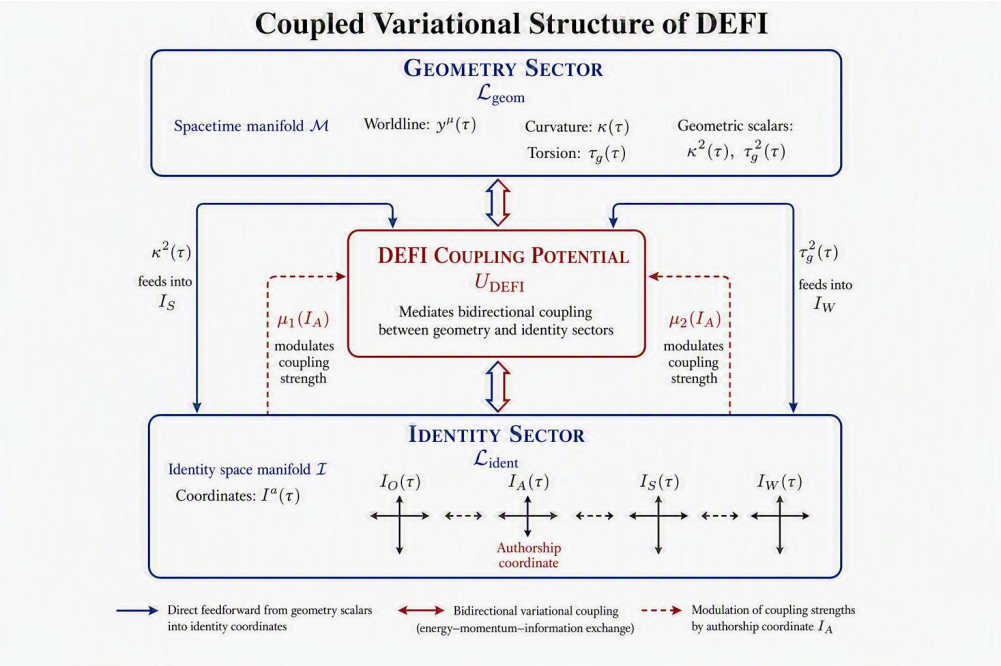


FIG. 3. Variational structure of DEFI. The coupling potential mediates bidirectional interaction between GWFM geometry variables and SEFI identity variables, with authorship-dependent coupling strengths.

- **Authorship as coupling weight, not an additive term.** In the original SEFI paper, Authorship is not its own geometric quantity but a weighted sum $F_A = \sum_i A_i(r,\kappa)$ over the other layers. Accordingly I_A appears here only through the coupling strengths $\mu_1(I_A), \mu_2(I_A)$, not as an independent term in U_{DEFI}.
- **Origin excluded.** I_O does not appear in U_{DEFI}, matching its role in the original SEFI paper as an *initial condition* rather than a dynamical potential term (Sec. VI).

V. THE DEFI ACTION AND EQUATIONS OF MOTION

The dynamics of S_{DEFI} follow from the standard Euler–Lagrange variational principle for a Lagrangian system on a product configuration space[4,8]:

$$S_{\text{DEFI}} = \int d\tau \left[L_{\text{geom}} + L_{\text{ident}} - U_{\text{DEFI}}(\kappa, \tau_g, I^a) \right]. \tag{7}$$

Varying with respect to the identity coordinates I_S, I_W gives geometry-driven identity dynamics:

$$\gamma\, \ddot{I}_S = -\mu_1(I_A)\, \kappa^2 - \gamma I_S, \tag{8}$$

$$\delta\, \ddot{I}_W = -\mu_2(I_A)\, \tau_g^2 - \delta I_W. \tag{9}$$

Varying with respect to I_A gives

$$\beta\, \ddot{I}_A = -\mu_1'(I_A)\, \kappa^2 I_S - \mu_2'(I_A)\, \tau_g^2 I_W - \beta I_A. \tag{10}$$

Varying with respect to y^μ (through κ, τ_g, functionals of y) feeds I_S, I_W back into the geometric equations of motion inherited from S_{WL} in GWFM, so that curvature and torsion

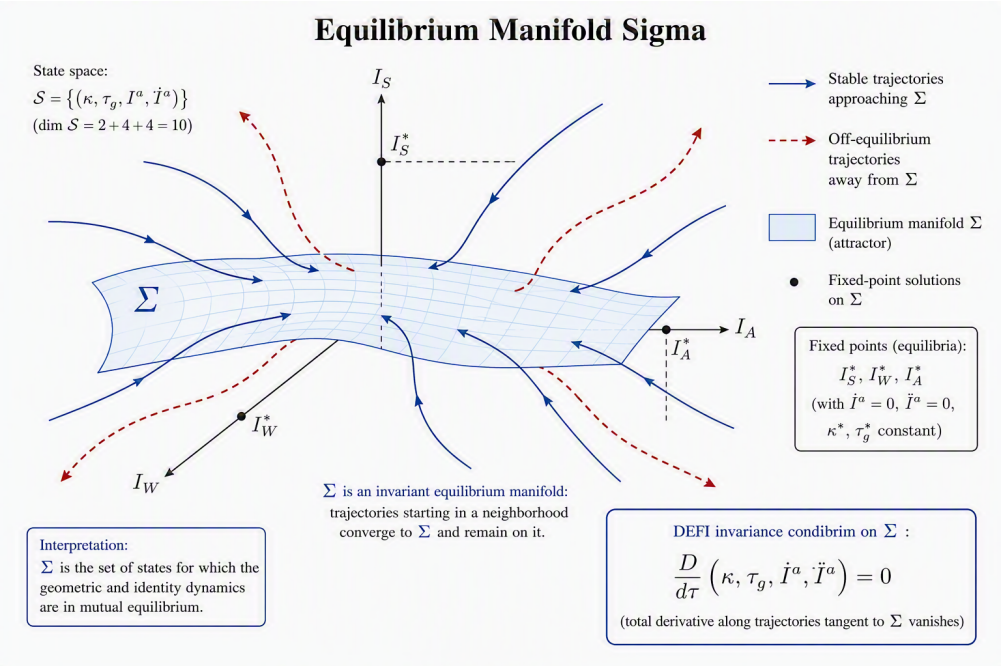


FIG. 4. Equilibrium manifold Σ defined by the fixed-point conditions of the coupled DEFI equations. Stable trajectories converge toward the manifold, while off-equilibrium trajectories move away from it.

no longer evolve freely but are pulled by the identity sector through the same potential U_{DEFI}. Neither sector is solved independently; both emerge as stationary points of the single action (7). This is the symmetric two-way coupling required of DEFI, obtained here as a derivation rather than an assumption.

A. Recovering the Stipulated Invariance Condition

A fixed point of the coupled system is a configuration at which every identity acceleration vanishes, $\ddot{I}_S = \ddot{I}_W = \ddot{I}_A = 0$, so that the identity sector is at rest relative to the instantaneous geometry (κ, τ_g). Note that this is *not* the same as setting $\partial U_{\mathrm{DEFI}}/\partial I^a = 0$: the harmonic terms of L_{ident} contribute to each equation of motion alongside U_{DEFI}, so the fixed point must be read off the full equations of motion of Sec. IV rather than from U_{DEFI} alone. Setting the left-hand sides of those equations to zero gives the fixed-point values explicitly, as functions of the instantaneous geometry:

$$I_S^*(\kappa, I_A) = -\frac{\mu_1(I_A)}{\gamma}\,\kappa^2, \tag{11}$$

$$I_W^*(\tau_g, I_A) = -\frac{\mu_2(I_A)}{\delta}\,\tau_g^2, \tag{12}$$

$$0 = \mu_1'(I_A)\,\kappa^2 I_S^* + \mu_2'(I_A)\,\tau_g^2 I_W^* + \beta I_A, \tag{13}$$

with Eq. (13) an implicit equation fixing $I_A^*(\kappa, \tau_g)$ once Eqs. (11)–(12) are substituted in. At any point $(\kappa, \tau_g, I_A^*, I_S^*, I_W^*)$ satisfying Eqs. (11)–(13) simultaneously, the identity accelerations vanish identically and the geometric equations of motion decouple from the identity sector at that instant. This is exactly

$$\frac{D}{d\tau}\left(\kappa,\ \tau_g,\ \dot{I}^a,\ \ddot{I}^a\right) = 0, \tag{14}$$

i.e., Eq. (1), now characterized as the fixed-point condition of the DEFI action rather than an axiom. The stability manifold Σ of the unified paper[3] is precisely the graph of Eqs. (11)–(13) in $M \times \mathcal{I}$ — the locus where the identity sector has relaxed onto the geometry it is coupled to — and the previously stipulated condition follows from Eq. (7) rather than defining it. Off this locus, Eqs. (11)–(13) fail to hold, $\ddot{I}^a \neq 0$, and the identity sector is out of equilibrium with the geometry, exactly as required for Σ to be a nontrivial proper subset of $M \times \mathcal{I}$ rather than the whole configuration space.

VI. CONSERVED ORIGIN CHARGE

Because I_O does not appear in U_{DEFI}, the Origin sector is fully decoupled: its equation of motion,

$$\ddot{I}_O = -I_O, \tag{15}$$

depends on no other variable in the theory. The Origin sector's own energy function,

$$Q_O = \frac{1}{2}\alpha\left(\dot{I}_O^2 + I_O^2\right), \tag{16}$$

satisfies $\dot{Q}_O = \dot{I}_O\left(\alpha\ddot{I}_O + \alpha I_O\right) = 0$ identically along the Origin equation of motion, independent of the state of the geometry or the other identity coordinates. This is the Noether charge[5,6] associated with the τ-translation symmetry restricted to the autonomous Origin subsystem: because L_{ident} contains no cross term coupling I_O to $(I_A, I_S, I_W, \kappa, \tau_g)$, the O-sector's own Hamiltonian is separately conserved even though it sits inside a larger interacting system. Q_O is therefore a genuine conserved quantity, not merely an initial condition — Origin is both where the trajectory starts and an invariant preserved throughout its evolution, consistent with its intended role as the foundational, unchanging layer of SEFI identity[1].

VII. RELATION TO GWFM, SEFI, AND THE UNIFIED THEORY

- **GWFM limit.** Setting $I^a = \text{const}$ (identity frozen) removes all coupling; $S_{\text{DEFI}} \to \int L_{\text{geom}}\, d\tau$, recovering pure GWFM worldline dynamics[2].
- **SEFI limit.** Setting $\kappa = \tau_g = 0$ (a straight, torsion-free worldline) removes the coupling term entirely; $S_{\text{DEFI}} \to \int L_{\text{ident}}\, d\tau$, recovering the identity-space dynamics of SEFI under Axiom 0's vectorization, with the original scalar F recovered by contraction (Sec. 1.1).
- **Unified operator.** The potentials $V_{\text{geom}}, V_{\text{ident}}, V_{\text{dyn}}$ of the unified transformation operator T^3 correspond respectively to the κ^2, τ_g^2 terms of L_{geom}'s curvature expansion, the quadratic terms of L_{ident}, and U_{DEFI} of Eq. (6). This construction therefore supplies the variational principle the unified paper's Limitations section identifies as missing.

VIII. LIMITATIONS AND FUTURE WORK

The coupling potential (6) is the simplest form consistent with the qualitative SEFI canon and full two-way symmetry; it is not derived from a deeper principle and other bilinear forms are equally consistent with the stated constraints. The functions $\mu_1(I_A), \mu_2(I_A)$ are unconstrained by the present construction beyond differentiability. Stability of the fixed-point locus Σ defined by Eqs. (11)–(13) (linear stability of the coupled equations about

that locus, in the sense standard to coupled-oscillator and dynamical-systems analysis[9]) has not been analyzed and is needed before Σ can be asserted to be attracting rather than merely stationary. Quantization of S_{DEFI}, and its compatibility with the eigenvalue problem $T\Psi_n = \lambda_n \Psi_n$ of the unified operator, is left to future work.

IX. CONCLUSION

DEFI has been constructed as a coupled Lagrangian dynamical system on $M \times \mathcal{I}$, with a fully symmetric coupling potential linking GWFM curvature/torsion to SEFI's Sovereignty and Warp identity coordinates, weighted by Authorship. The previously stipulated dynamical invariance condition is recovered as an equilibrium condition of this system rather than assumed, and a conserved Origin charge has been derived via Noether's theorem for the decoupled Origin sector. This construction required an explicit, stated extension of SEFI's scalar identity functional to a four-vector, recorded here as Axiom 0 rather than left implicit.

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